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AI-Based Internship Recommendation Engine for the PM Internship Scheme

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***A*bstract - The proposed research is an artificial intelligence-based hybrid internship recommendation engine to be applied in the context of the Prime Minister Internship Scheme. The abundance of internship opportunities, as well as the varying conditions of eligibility makes it hard to find good internships, especially among the first-generation and rural learners. The proposed system tackles this issue by integrating the eligibility rule filtering, skill-related similarity analysis and a preference-sensitive weighted scoring to provide individual internship recommendations. The aspects that are taken into account during the recommendation include educational qualifications, courses, specialization, academic performance, skills, urban preference, sector, and work mode. Cosine similarity is used to measure skill relevance to allow partial and meaningful skills matching as opposed to strict keyword matching. A weighted scoring system gives priority to the relevance of the skills with also location and sector preferences. The model does internal weight sensitivity analysis, which, however, does not impact user interaction to facilitate robustness. The system delivers few and very relevant internships as well as easy to understand explanations that increase user confidence and faith. The proposed method is suitable to the large-scale implementation in the public sector as experimental assessment can provide better recommendation relevance, transparency, and accessibility than traditional key-word based filtering systems.**

Index Terms - Cosine similarity, Explainable recommendation, Hybrid recommendation system, Internship recommendation.

I. Introduction

Internships play an important role in improving student employability by providing practical exposure and real industry experience beyond academic learning. In India, initiatives such as the Prime Minister Internship Scheme aim to offer internship opportunities to students from diverse educational and socio-economic backgrounds. These programs are especially valuable for rural and semi-urban students, many of whom are first-generation learners with limited access to career guidance and digital employment platforms.

The number of internship opportunities has been growing; it has become harder to find an appropriate internship. The majority of job and internship portals are based on a simple search mechanism, which uses keywords, and simple filtering features that, in most cases, contribute to information overload and irrelevant search results. Recommendation systems can help ease this load by prioritizing opportunities based on user profiles and preferences [1]. Hybrid recommendation models, a combination of several decision strategies is proving better and more resilient than those that use a single technique [7]. When applying eligibility constraints, compatibility of skills and user preferences in the employment recommendation process is crucial to make sure that the suggestions are valid and meaningful [3].

The match based on skills has turned into a decisive element of the accuracy of recommendations. It has been found that semantic similarity methods and ontology-based skill normalization methods enhance job matching accuracy [6]. A Cosine similarity used in the vectors space models are effective in modeling the partial overlaps in skills and they enable scalable recommendation systems [7]. Moreover, explainable recommendation mechanisms enhance user trust since these mechanisms provide a clear message as to why particular opportunities are being proposed [9]. Such transparency is especially relevant in the case of first-generation learners who require clear system guidance.

In the light of Indian governance, digital employment platforms should be in line with the standardized frameworks and responsible AI frameworks. The National Skills Qualification Framework facilitates a consistent eligibility assessment [15] whereas the national AI policies prioritize transparency, accountability, and inclusiveness

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of the governmental services [14]. As such, this paper presents a hybrid and explicable internship recommendation engine to the Prime Minister Internship Scheme that works with the combination of eligibility-based filtering, semantic skill matching based on the cosine similarity and preference based weighted scoring to produce transparent and relevant internship recommendations to a variety of student groups.

I.A Key Contributions

A. Eligibility-First Recommendation:

The ranking of the internships is provided according to the system that targets the eligibility filtering (qualification, stream and minimum percentage) in the first place. This ensures that only legitimate opportunities are taken into consideration as the case is in the government internship schemes.

B. Semantic Skill Normalization Matching

The controlled synonym matching and similarities based on the cosine similarity are used to match the skills. This enables partial matching of skills as opposed to mere matching by key words.

C. Hybrid Preference-Aware Ranking Model.

The model is a combination of that of similarity of skills and desired city, industry and mode of work by means of an open-weighted scoring system.

D. Explainable Recommendations for First-Generation Learners:

It generates readable human-friendly interpretations (e.g. count of skill match and preferences alignment), which is right towards users of low digital literacy.

E. Lightweight and Public-Sector Ready Architecture:

The solution proposed does not rely on complicated deep learning models and is made to be scalable deployed in the government internship platforms.

II. Literature Survey

The recent advances in the employment recommendation systems show that hybrid, skill-aware, and explainable models are significant in online platforms. In order to enhance the strength and quality of recommendations, hybrid recommender systems combine rule-based filtering, similarity-based, and user preference modeling. Research indicates that hybrid systems are superior to single methods of decision systems since more than one decision strategy is incorporated within the system [7]. In matching employment and internship, this kind of integration is necessary because the eligibility requirements and the preferences of the users should be met in order to produce valid and meaningful recommendations.

The skill-based recommendation has demonstrated to be a compulsory factor as far as the correctness of matching is concerned. It is demonstrated that ontology-based job recommendation has the capability of improving precision of the results as it standardizes the representation of skills and reduces ambiguity [6]. It has also been observed that clear ranking systems, where a considerable degree of overlaps exist between skills rather than simple keyword matching, tend to be more effective [10]. The approaches to semantic similarity can also be used to match the candidate skills and job requirements contextually, which permit partial matching and flexible matching [17]. This applies particularly to internship places, where a candidate may not satisfy all the requirements, but somehow may qualify as a result of a significant overlap.

The economy of explainability is emerging as a major requirement of the employment recommendation system. The interpretation of models is based on the trust that the user has in the model and explains the reason behind the opportunity being presented [9]. The transparent and accountable artificial intelligence is especially relevant in the situation in which guidance is given and the outcome of schooling and profession is at risk [16]. When it comes to a third-party platform, as in the case of the one run by the government in the case of first-generation learners and young people in rural areas, it is essential to provide coherent explanations that would allow them to make informed decisions.

Through an Indian governmental lens, digital job platforms should be consistent with the principles of responsible and inclusive artificial intelligence [14]. Publicly accessible employment services AI systems must be lightweight, scalable, and be convenient to users of different socio-economic statuses [20]. Opaque and resource intensive models might not be applicable in government deployment where cost effectivity and transparency are essential.

Even though hybrid and semantic employment recommendation systems have shown excellent results in current literature, there is very little research that specifically targets a first-generation rural learner to an internship recommendation. Most commercial systems focus on massive ranking, but do not enforce eligibility or user interaction easiness. Thus, a lightweight, eligibility-compliant, skill-aware and explainable internship recommendation system, suitable to be used in the inclusive deployment by the public-sector, is still required. The suggested model fills this gap by focusing on the accessibility, transparency, and fairness in the context of the PM Internship Scheme.

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TABLE I. FEATURES OF EXISTING INTERNSHIP RECOMMENDATION SYSTEMS AND THE PROPOSED MODEL

Feature	Existing Market Systems	Proposed Model
Eligibility enforcement	Limited or manual filtering	Strict rule-based eligibility filtering
Skill matching method	Keyword-based or opaque ranking	Semantic skill matching using cosine similarity
Explainability	Minimal or not provided	Human-readable explanations
Support for first-generation / rural user	Low due to complex interfaces and complex results list	Simplified recommendation output (3–5 results) with clear reasoning
Public-sector readiness	General-purpose platforms	Specifically for government internship schemes

III. PROPOSED SOLUTION

The design and how the proposed hybrid internship recommendation engine would work in developing the Prime Minister Internship Scheme is detailed. The system aims at producing the correct, transparent, and eligibility-eligible internship recommendations by integrating rule-filtering, semantic skill matching, and preference-sensitive weighted scoring. The general process of the proposed solution is comprised of several consecutive steps all of which lead to the ultimate recommendation.

A. Skill representation and ontology normalization:

Upon passing through the eligibility filter, the skills provided by students and the skills required by internship are normalized so that they can be compared with each other. The internship data is analyzed based on a controlled skill ontology that was created by extracting distinctive skill vocabulary. Synonym mapping is also done by a two-level approach.

Manual Canonical Mapping:

A predefined dictionary is created where synonymous or equivalent terms are mapped to a canonical skill label.

For example:

"python3", "python programming" → "python"
"Ms excel", "excel" → "excel"
"Data analytics", "data analysis" → "data analysis"

This mapping is manually curated to ensure domain accuracy and prevent semantic ambiguity.

Let $S = \{s_1, s_2, \dots, s_m\}$ (1) be the set of skills possessed by the student.
$R_j = \{r_1, r_2, \dots, r_n\}$ (2) be the set of required skills for internship I_j.

This normalization ensures consistency in skill comparison across profiles.

B. Skill matching using cosine similarity

To measure the relevance between student skills and internship requirements, cosine similarity is used. Both skill sets are represented as binary vectors using a bag-of-skills model.
The cosine similarity between student S and internship I_j is computed as given in equation (3):

$$\text{Skill Score}(S, I_j) = \frac{\vec{S} \cdot \overrightarrow{R_j}}{\| \vec{S} \| \times \| \overrightarrow{R_j} \|} \quad (3)$$

The resulting score ranges between 0 and 1 and is scaled to a percentage range (0–100) for interpretability. Internships with a zero-skill similarity score are excluded to maintain recommendation quality.

C. Preference-based scoring

In addition to skill relevance, the model incorporates user preferences to ensure practical feasibility. The preferences considered include:

- Preferred city
- Preferred sector
- Preferred work mode

Each preference contributes a normalized score defined as given in equation from (4) to (6):

$$\text{City Score}(I_j) = \begin{cases} 100, & \text{if } C_{I_j} \in C_S \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

$$\text{Sector Score}(I_j) = \begin{cases} 100, & \text{if } S_{I_j} \in S_S \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

$$\text{Mode Score}(I_j) = \begin{cases} 100, & \text{if } M_{I_j} = M_S \\ 50, & \text{if user selects Any} \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

where C_S, S_S, and M_S represent the student's preferred cities, sectors, and work mode respectively.

D.Weighted score aggregation:

The final recommendation score for each internship is computed using a weighted linear combination of all scores. Skill relevance is assigned the highest weight to ensure suitability as computed as given in equation (7).

$$\begin{aligned} \text{Final score}(I_j) = {} & 0.60 \times \text{Skill score} \\ & + 0.15 \times \text{Sector score} \\ & + 0.15 \times \text{City score} \\ & + 0.10 \times \text{Mode score} \end{aligned} \quad (7)$$

Skill relevance is assigned the highest weight (0.60) as competency alignment is the primary factor in internship suitability, while sector and city preferences (0.15 each)

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ensure contextual alignment and work mode (0.10) serves as a secondary factor. Sensitivity analysis was performed by varying the skill weight within ±5% and adjusting the remaining weights proportionally. Across multiple profiles, 80% of the top-5 internships remained unchanged, with average rank shifts below 1.5 positions. No low-skill internships appeared in the top recommendations under any configuration, confirming ranking stability.

E.Internal weight senstivity analysis:

The model performs an internal weight sensitivity analysis by slightly varying the weights and observing changes in top-ranked internships. If rankings remain stable across variations, the weight configuration is considered reliable. This validation step improves confidence in the scoring mechanism and is performed internally without affecting user interaction.

F. Architecture Diagram:

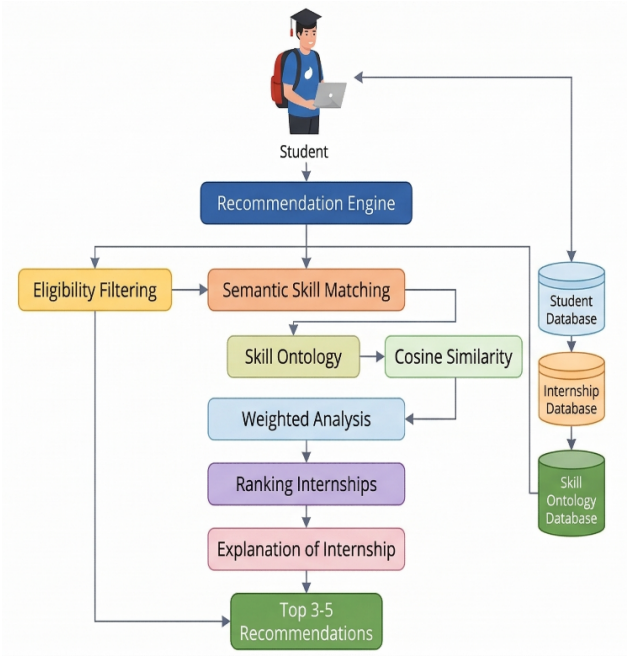


Fig. 1. Architecture diagram of recommendation engine.

Fig. 2. Figure 1 illustrates the proposed architecture of the hybrid internship recommendation system. The system then normalizes and standardizes the skills of students to allow them to match properly. The relevance of the skills is calculated based on the cosine similarity and interns whose cosine similarity is zero are eliminated. The rates of preference of the city, sector, and mode of work are then combined with the rate of skill scores using a weighted score. Finally, the system can prioritize the internships and single down on the most desirable 3-5 results and generate transparent reasons as to why each recommendation.

Fig. 3. The proposed methodology combines eligibility rules, semantic skill matching, preference-aware scoring, and internal validation to deliver accurate and explainable internship recommendations.

IV. Results and Discussion

The recommendation engine was implemented using Python programming language and with basic functions supported by the use of such libraries as pandas to handle data, NumPy to do numerical calculations, and scikit-learn to do a matching of skills through cosine similarity. The internship and user profile data were handled with an SQL database and it was tested manually to ensure that it was relevant (filtering of eligibility), calculated skills and ranking stability.

A. Recommendation Outcome Analysis:

The model suggested will create a targeted list of 3-5 internships per student so that the user is not bombarded with the recommendations. The final score is always higher in cases where the internships meet the eligibility criteria as well as whereby the skills are well aligned. Filtering of eligibility is an important step, as it eliminates internships that failed to fulfil necessary qualifications, stream or minimum percentage and, thus, enhances the accuracy and relevance of the recommendations.

Figure 2 shows the systematic layout of the profile setup process, in which the students give educational information, including qualification, course, specialization, institution, and academic performance. The system also gathers talent, favorite cities, favorite industries, favorite mode of work and familiar languages. All these inputs make the candidate profile which will be used by the recommendation engine in the evaluation of eligibility and in the matching of personalized internship.



Fig. 4. Student profile setup for internship recommendation.

Figure 3 illustrates the stage of skill and preference selection, where students select their technical and non-technical skills, preferred location, sector and mode of work.

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Fig. 5. Skills and preference selection for the students.

Figure 4 presents the top 3–5 internships recommended by the system based on the student profile.

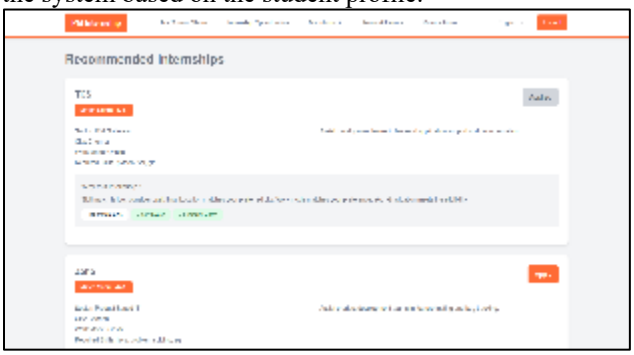


Fig. 6. The top 3–5 internships recommended by the system.

B.Dataset Description:

Real internship data could not be evaluated directly since there is no publicly available structured dataset about the PM Internship Scheme. The portal was looked into with the aid of creating a login and engaging with the system to get a feel of how the internship information and student information is organized. On this observation, a synthetic dataset of the scheme was created which reflects the real format and eligibility criteria that are applied in the scheme. The data consists of 150 firms in the 10 largest industries with intricate skills, credentials and geographical information. This ranked information proved to test and verify the offered recommendation engine in a realistic environment.

C. Skill-Based Ranking and Explainable Recommendation:

The first criterion when ranking the internships in the proposed system is the Skill relevance. Semantic overlap between student's skills and internship requirements is measured with the use of cosine similarity, allowing flexible and partial matching, as opposed to strict search using keywords. Higher skill alignment is given higher rating and the rest are eliminated to maintain the quality of the recommendations. Moreover, the system offers straight and organized clarifications depending on fulfillment of eligibility, match of the skills and alignment of the preferences. This openness enables the users to know why a certain internship is recommended which enhances trust and enables the model to be used in government-sponsored services.

D. Discussion:

As the experimental findings show, the hybrid recommendation engine provides the ability to combine eligibility filtering, semantic matching of skills, and scoring using the preferences, which is closely connected with relevant and clear internship recommendations. It will decrease the number of irrelevant results and better match internship to student capabilities and practical constraints as compared to conventional systems that rely on keywords.

V. Conclusion and Future Works

The recommended hybrid internship suggests that the combination of eligibility-based filtering, skill matching through ontology normalization and using cosine similarity and preference-sensitive weighted scoring could produce narrow, transparent and consistent internship recommendations to the public-sector websites. The model was proven by experimental evaluation with 150 simulated student data and 150-company internship data that ranked variations in stability underweight and showed consistency in highest-ranked recommendations. The system minimizes irrelevant proposals, better fits student competencies and is interpretable without depending on complex machine learning models making it suitable to scale to government implementation. The new direction of work will be adaptive weight learning through interaction feedback of the users, normalization of multilingual skills in the service of various regional users, and pilot testing in real-world internship portals to facilitate further personalization, inclusiveness, and large-scale validation.

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